# PERMUTATIONS & COMBINATIONS-I

1. A coin is tossed 3 times and the outcomes are recorded. How many possible outcomes are there?
2. A dice is tossed 4 times and the outcomes are recorded. How many possible outcomes are there?
3. In how many different ways letters of the given word be arranged.
   (i) MATHS (ii) DISPLAY (iii) MACHINE (iv) EDUCATION
   (v) OFFICERS (vi) MATHEMATICS (vii) EXAMINATION
4. How many ways the following words can be arranged
   i) Repetition allowed ii) Repetition not allowed
   1) PIRATES 2) CRYSTAL 3) BOOK 4) INSTITUTE
5. In how many ways can the letters of the word UNIVERSAL be arranged when E,R,S are always together?
6. How many ways letters of the given word be arranged vowels always comes together
   (i) JUDGE (ii) CODEGNAN (iii) HYDERABAD (iv) VIJAYAWADA (v) OPERATE
7. How many ways letters of the given word be arranged consonants always comes together
   (i)INFORMATION (ii) BANKER (iii) CORPORATION
8. How many ways letters of the given word be arranged vowels never comes together
   (i) SISTER (ii) BROTHER (iii) EXAMINATION
9. In how many ways can the letters of the word ALGEBRA be arranged when both A do not come together?
10. Number of ways of permuting the letters of the word "**ENGINEERING**" so that the order of the vowels is not changed is:
11. How many ways the word **JOURNALISM** can be arranged
    i)If the word starts with letter 'R'?
    ii)If the word starts with letter 'R' and ends with letter 'N'?
    iii)The word should always ends with Vowel?
    iv)The word starts with a vowel and ends with a consonant?
    (v) No two vowels comes together

## PRACTICE QUESTIONS:

1. Find the number of 3 letter words, with or without meaning, which can be formed out of the letters of the word DAD, where repetition is not allowed?
2. In how many different ways can the letters of the word BELIEVE be arranged?
3. Find the number of permutations of the letters of the word ALLAHABAD?
4. In how many ways the letters of the word SACRED can be arranged so that vowels come together?
5. In how many of the distinct permutations of the letters in MISSISSIPPI do the four I's come together?
6. Find the number of 4 letter words, with or without meaning, which can be formed out of the letters of the word ROSE, where repetition of the letters is not allowed?